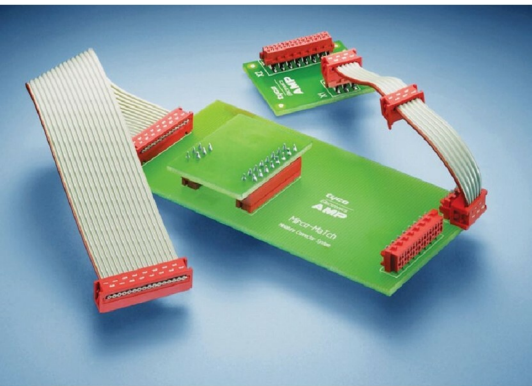


How to Choose the Right Connector

As it is important to choose the right connector for an embedded system, let us share some valuable advice from industry professionals



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A connector may seem to be a relatively unimportant component when we first look at any device. But design engineers know the importance of choosing the right connector for their design. A connector is the base of an electronic product, as it binds the entire system together. Also, it plays a key role in the successful design and reliable operation of an embedded system.

Interestingly, knowing better about the connectors enables us to choose an ideal match for our application. It is easy to design an optimal product by doing this in the early stages of an embedded system design process. It could make our system smaller, lighter and easier to handle. It also helps reduce cost and increase usability.

How connectors have changed

Just like other electronic components, connectors are also available in thousands of variants. Specifications and dimensions range from connectors for the smallest IC to big power connectors carrying very high current values on board.

With respect to the embedded connectors, a lot of changes have been witnessed over a couple of years. And on the design side, a vast majority of modifications have been implemented in response to consumer electronics trends.

Symmetry. The trend of symmetrical connectors was started by Apple, with the creation of lighting connectors for their branded portable devices. This trend has seemingly caught on. Apple has recently announced a new

USB type-C connector, which is a reversible variant, which means there is no 'wrong' way to connect it. You can connect it upside down and it will still work.

Miniaturisation. The ongoing requirement for portable equipment is continuing to drive the general miniaturisation trend. To remain competitive and cope with an increasing demand for smaller circuits, connector manufacturers have further decreased the size of their connectors.

But despite their smaller size, the miniaturised connectors are engineered to still be able to pass large currents that were previously handled by their bigger predecessors. However, smaller electromechanical components have a problem of breaking easily.

The line dividing smaller connectors and fragile connectors. Manufacturing technology evolved to accommodate ruggedness to tackle the problem of fragility with reduced size. For instance, the housings are made from heat-resistant, self-extinguishing thermoplastic material in cases where environmental sealing is required.

More robust connectors can easily resist very harsh environments for a longer period of time. Considering the ease of servicing, pluggable connectors with higher ingress protection (IP) rating and better locking features have also come up. The connectors are also designed to be easier to connect, using technology such as 'push-in connection' that gives the connector an integrated spring lever.

Connectors also have features implemented to improve stability. For instance, TE Connectivity's Power Triple Lock line features two optional lock—connector position assurance which